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A real-time decision-making and motion planning framework with collision avoidance for intelligent vehicle

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***Abstract*—Safety and real-time performance are key requirements for decision-making and motion planning in intelligent vehicles. However, existing learning-based methods lack interpretability in terms of safety, making it difficult to effectively avoid collision risks. Additionally, rule-based approaches face challenges such as planning failures and limited real-time performance. To address these issues, an integrated framework for real-time decision-making and motion planning focused on collision avoidance is proposed. First, in terms of safety, a vehicle motion-based safety assessment module (VMSA) is developed to provide interpretable safety guarantees. This ensures continuous checking and timely adjustment of lane change gap, effectively preventing planning failures. Second, in terms of real-time performance, a motion planning module based on multi-objective deep reinforcement learning (DRL) and transfer learning is designed to optimize motion trajectory generation. Additionally, a DRL policy based on global traffic features is proposed to optimize lane selection, and curriculum learning is employed to enhance training stability. Simulation results show that, compared with other methods, the proposed framework enables real-time decision-making while effectively improving driving efficiency and safety.**

***Index Terms*—Motion planning, collision avoidance, real-time, intelligent vehicle, deep reinforcement learning.**

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I. Introduction

With the advancement of artificial intelligence and urbanization, autonomous driving technology has been widely studied to improve driving safety, traffic efficiency, and convenience [1–3]. Autonomous systems mainly consist of perception, decision-making, motion planning, and control modules [4, 5]. As core components, the decision-making and motion planning modules are typically studied independently, determining the vehicle's actions and paths in specific scenarios. However, achieving efficient autonomous driving in the future requires a control approach that deeply integrates decision-making and motion planning while ensuring safety and real-time performance.

Lane change is a typical application scenario for decision-making and motion planning modules [6–8]. Proper lane changes can improve traffic efficiency and road utilization. However, the lane change process poses several challenges. First, the presence of human-driven vehicles, non-motorized vehicles, and pedestrians on the road increases safety risks [9]. Second, real-time responsiveness is important, enabling the adaptation of driving behaviors promptly to changing traffic conditions [10]. These challenges make lane change a central issue in the research of autonomous driving decision-making and motion planning.

To date, many studies have focused on safe and efficient lane change. The main approaches can be categorized into four types: sampling-based methods, methods that separate decision-making and motion planning, integrated methods, and learning-based methods.

Sampling-based methods select a trajectory from a set generated by rule-based models connecting start and target points. Kou et al. [11] used sextic polynomials to generate

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efficient lane-change trajectories. Sun et al. [12] applied a third-order minimum curvature method for energy-efficient planning. However, as it does not consider interaction with the environment, it is primarily used for basic control tasks. Moreover, generating many candidate trajectories before each lane change leads to high computational cost.

In certain special cases, relying solely on the planning module is infeasible. For example, when there is no safe gap for lane change in the target lane, forcing a lane change could result in collisions. Therefore, considering interaction with the environment is essential. Most existing studies address decision-making and planning modules separately. Decision-making methods include game theory, Partially Observable Markov Decision Processes (POMDP), and learning-based approaches [13–15]. Planning methods involve graph search, interpolated curves, sliding-mode control and learning-based algorithms [16–18]. While this separation is simple and intuitive, it has two main drawbacks. First, planning results are not promptly fed back to the decision-making module, hindering policy performance. Second, decision-making often prioritizes immediate gains, risking locally optimal rather than globally optimal solutions.

Recently, several studies have proposed integrated methods. Based on different decision-making approaches, these methods can be categorized into three types: risk-based analysis, game theory, and learning-based approaches.

Risk-based analysis models the risk using temporal and spatial relationships between vehicles and the uncertainty of input data to minimize risk in driving decisions [19]. For example, Wang *et al.* [20] proposed a motion planning framework that uses a risk model based on distances and Model Predictive Control (MPC) to minimize risk and trajectory deviation. To avoid collisions caused by mandatory lane changes, Wang *et al.* [21] developed a risk model based on the Artificial Potential Field (APF) theory. Game theory-based methods analyze competitive or cooperative relationships between lane change vehicles and surrounding vehicles to identify optimal decisions. For example, Pan et al. [22] combined game theory with Model Predictive Control (MPC) to improve lane change safety. Zhao et al. [23] adopted a Stackelberg game and social potential fields to design a human-like decision controller. Learning-based methods optimize policies by interacting with the environment [24]. Liao et al. [25] proposed a DQN-based decision module for collision-free decisions, and an MPC-based planner for comfortable trajectories. While integrated approaches enable lane changes in specific scenarios, most rely on MPC for trajectory planning. This poses challenges for real-time use due to the high computational cost of solving nonlinear problems, such as the 24 seconds reported in Ref. [25]. In addition, lane change failures from poor trajectory planning have also been reported [26].

Learning-based methods typically use deep reinforcement learning (DRL) to directly map state observations to action outputs. Depending on the type of action space, these methods can be categorized into discrete and continuous actions [27–30]. For instance, Gao *et al.* [31] used DRL to decide the lane for the vehicle in the next time step, thereby improving traffic efficiency. In Ref. [32], the soft actor-critic algorithm was used as the decision method to achieve conservative and aggressive driving strategies, where the action space consists of continuous steering angles and acceleration. Li *et al.* [33] proposed a risk-aware strategy to ensure safe lane changes. This strategy evaluates risks using a probabilistic model, and employs DRL to learn risk-minimizing policies. Learning-based methods ensure real-time decision-making but suffer from poor interpretability, making it challenging to meet safety requirements.

The mainstream methods are integrated methods and learning-based methods. Integrated methods ensure safety, but most of the planning modules rely on MPC. This approach is not conducive to the real-time performance of autonomous driving. Moreover, most existing integrated methods separate decision-making from motion planning, leading to the potential issue of planning failure. Learning-based methods mainly rely on DRL, but policies trained solely with collision penalties lack interpretability and perform poorly in avoiding collisions (Tables III and IV). Inspired by the strong interpretability of integrated methods and the high real-time performance of learning-based methods, this paper proposes a real-time integrated framework with collision avoidance for autonomous driving decision-making and motion planning, as shown in Fig. 1. Specifically, a trajectory planning network is designed with consideration for multi-objective optimization, meeting the real-time requirements. Additionally, a vehicle motion-based safety assessment module (VMSA) is constructed to achieve collision avoidance. Finally, lane selection based on global traffic features is implemented, improving driving efficiency.

The main contributions are summarized as follows.

1) A novel integrated decision-making and motion planning framework is proposed to reduce inconsistencies between modules.
2) A vehicle motion-based safety assessment module (VMSA) is introduced. It ensures policy safety and interpretability by explicitly modeling safety constraints.
3) DRL policies driven by global traffic features and multiple objectives are designed for real-time lane selection and trajectory planning.

This study hypothesizes that by introducing the VMSA module and designing lane selection and motion planning policies based on DRL, intelligent vehicles can ensure safety and enhance real-time performance in mixed traffic scenarios. Specifically, we predict that the proposed integrated framework can effectively avoid collision risks and improve driving efficiency through real-time policies.

The structure of the rest of the paper is as follows: Section II presents the system model. Section III designs the proposed integrated framework, including the motion planning module, safety assessment module, and lane selection module. Section IV provides experimental validation in multiple scenarios. Finally, the conclusion is given in Section V.

II. System model

A. Problem statement

Considering the general applicability of the scenario, a multi-vehicle scenario on a three-lane road was chosen. There

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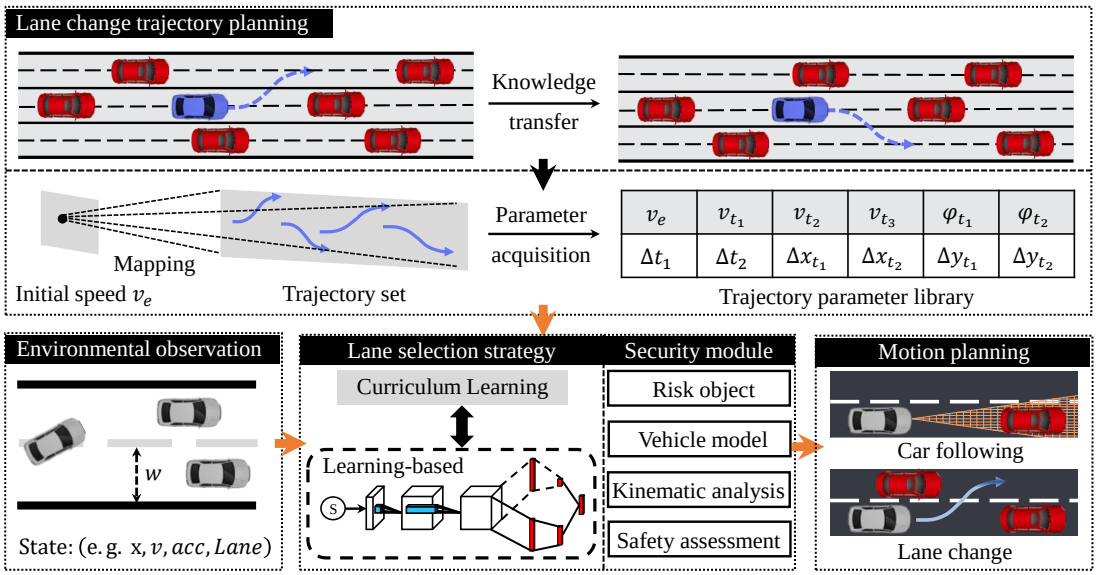

Fig. 1. The overall structure of the proposed integrated framework.

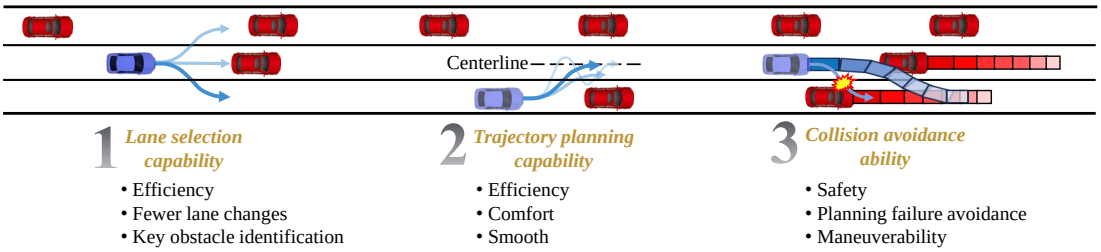

Fig. 2. Schematic diagram of the three focused objectives.

is an intelligent vehicle and several human-driven vehicles evenly distributed on the road.

When the intelligent vehicle (the controlled vehicle) is blocked by an obstacle ahead, it should decide whether to continue in the current lane or change to another lane. The goal is to explore a policy that enables the intelligent vehicle to safely and efficiently change lanes in mixed traffic, while improving driving efficiency. Specifically, this can be divided into three sub-goals: lane selection, trajectory planning, and collision avoidance, with integrated module coupling, as shown in Fig. 2. Factors such as measurement errors [34], delays [35], and model complexity can affect the performance of autonomous systems. To better focus on the research of the core framework, the following assumptions are made:

1) The controlled vehicle can accurately perceive the necessary environmental information.
2) The motion of surrounding vehicles is modeled using the Intelligent Driver Model (IDM), excluding special cases such as rear-end collisions caused by inattentiveness of following vehicle drivers.
3) To simplify interaction modeling, lane changes are allowed only when surrounding vehicles move straight; otherwise, the controlled vehicle follows IDM to avoid uncertain collision risks.
4) The vehicle motion is simplified to the two-degree-of-freedom (2-DOF) model.

The 2-DOF kinematic model is used to describe the vehicle state transitions, as follows:

$$\begin{bmatrix} x' \\ y' \\ \varphi' \\ v' \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & \sin(\beta+\varphi) \\ 0 & 0 & 0 & \cos(\beta+\varphi) \\ 0 & 0 & 0 & (\sin\beta)/l_r \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ \varphi \\ v \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} a \tag{1}$$

where $\bullet'$ represents the derivative of state parameter $\bullet$. x and y are the vehicle's position parameters, representing the longitudinal and lateral positions of the center of mass, respectively. v is the speed. and φ is the heading angle. β is the slip angle at the vehicle's center of mass. l_r is a vehicle structural parameter, denoting the distance from the center of mass to the rear axle.

B. Surrounding vehicle behavior controller

By using a controller to manage the longitudinal motion, the behavior of drivers in human-driven vehicles can be simulated. In this study, IDM is employed as a controller to adjust speed and facilitate interaction with the intelligent vehicle [36].

The acceleration a_t obtained at each moment can be expressed as:

$$a_t = a_{\max} \cdot \left[1 - \left(\frac{v}{v_d} \right)^{\lambda} - \left(\frac{d_d}{d} \right)^2 \right] \tag{2}$$

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where $a_{\max}$ is the maximum acceleration of the controlled vehicle. v_d is the desired speed, which is typically set to the maximum speed. d denotes the longitudinal distance between the controlled vehicle and the preceding vehicle in the current lane. d_d is the desired distance, which can be calculated based on the speed and acceleration as follows:

$$d_d = d_0 + T_0 \cdot v + \frac{v \cdot \Delta v}{2\sqrt{a_{\max} \cdot b_{\max}}} \tag{3}$$

where d_0 is the minimum safety distance. T_0 is the desired time headway, and $b_{\max}$ is the maximum deceleration.

III. Integrated Framework

This section introduces the proposed safety-interpretable integrated framework for autonomous driving decision-making and motion planning. The lane change task is decomposed into lane selection and motion planning, with dedicated modules designed for each task. Additionally, a VMSA-based safety assessment module is proposed to enhance the safety of decision-making and the success rate of lane changes.

A. Motion planning module

This subsection describes the design of the motion planning module. A simple cruise control module is used for longitudinal motion. For lateral motion, the TD3 algorithm combined with transfer learning (TL), referred to as TTL, is used to generate lane change trajectories with high real-time performance.

1) Problem modeling: We use RL to model the motion planning problem. The agent learns useful strategies through interaction with the environment and reward feedback. This process is a Markov Decision Process (MDP). At each time step, the agent collects the observation s_t through sensors and selects an action a_t based on the policy π. After executing the action, the agent receives the corresponding reward r_t and the next state s_{t+1}.

The state space is designed to reflect key states with minimal dimensional information. For the lane change task, the state space includes observations of both the ego vehicle's state and the environment. The lateral deviation Δy between the vehicle and the target lane centerline is used to represent the spatial relationship. To describe other vehicle states, the state space includes heading angle φ, speed v, and acceleration a. The state space S is defined as:

$$s_t = (\Delta y(t), \varphi(t), v(t), a(t)) \in S \tag{4}$$

The action space includes continuous steering angle δ_f and acceleration a to reflect realistic dynamics, defined as follows:

$$a_t = (\delta_f(t), a(t)) \in A \tag{5}$$

The reward function plays a crucial role in RL, directly influencing the success of the training process. To achieve efficient and comfortable lane change, the following factors are considered to evaluate the actions:

- Accuracy: changing to the target lane centerline and driving along it.
- Efficiency: maximizing driving speed within the speed limit.
- Comfort: keeping low acceleration to enhance the ride experience.
- Smoothness: keeping low yaw rate acceleration for smooth lane change.
- Penalty: applying a huge penalty for failures to accelerate training.

The immediate reward r_t obtained by the agent at each time step is as follows:

$$\begin{aligned} & r_t = r_{lane} + r_{speed} + r_{comfort} + r_{smooth} + r_{fail} \\ \text{s.t.} \quad & r_{lane} = \frac{w_1}{1 + w_2 \left|\Delta y\right|} \\ & r_{speed} = w_3 \frac{v - v_{\min}}{v_{\max} - v_{\min}} \\ & r_{comfort} = -w_4 a^2 \\ & r_{smooth} = -w_5 a_{\text{yaw}}^2 \\ & r_{fail} = \begin{cases} -w_6, & \text{if fail} \\ 0, & \text{else} \end{cases} \end{aligned} \tag{6}$$

where w_i is a coefficient balancing the rewards. Among them, $w_1 = 6, \quad w_2 = 5, \quad w_3 = 1.5, \quad w_4 = 0.2, \quad w_5 = 0.1, \quad w_6 = 60$. v_{max} and v_{min} are the maximum and minimum speeds. a_{yaw} is the yaw rate acceleration. r_{lane} encourages minimal lateral deviation from the target lane, while also improving lane change efficiency like r_{speed}. $r_{comfort}$ and r_{smooth} focus on trajectory comfort and smoothness. r_{fail} helps differentiate successful and failed trajectories to speed up training.

2) TTL-based solution: TD3 is widely used for handling problems in continuous action spaces. In this algorithm, the agent selects actions based on the actor network:

$$a_t = \pi_\phi(s_t) + N_t \tag{7}$$

where ϕ represents the parameters of the policy network. N_t is the exploration noise to stabilize the policy and avoid local optima. The interaction experience is stored in the replay buffer as a tuple $(s_t, a_t, r_t, s_{t+1}, d_t)$ for training, where d_t indicates the episode's termination.

During training, N samples are randomly sampled to compute the Q-value y of the target critic network, as shown below:

$$y = r + \gamma \min_{i=1,2} Q_{\theta_i'}(s', \tilde{a}) \tag{8}$$

where $Q_{\theta_i'}$ is the parameter of the i-th target critic network, s' is the next state in the sample, and $\tilde{a}$ is the action selected based on the target actor network with random noise.

The critic network is updated by minimizing the loss. The loss L_i is calculated as follows:

$$L_i = \frac{1}{N} \sum \left(y - Q_{\theta_i}(s, a)\right)^2 \tag{9}$$

where L_i and Q_{θ_i} represent the loss and the Q-value of the i-th critic network, respectively.

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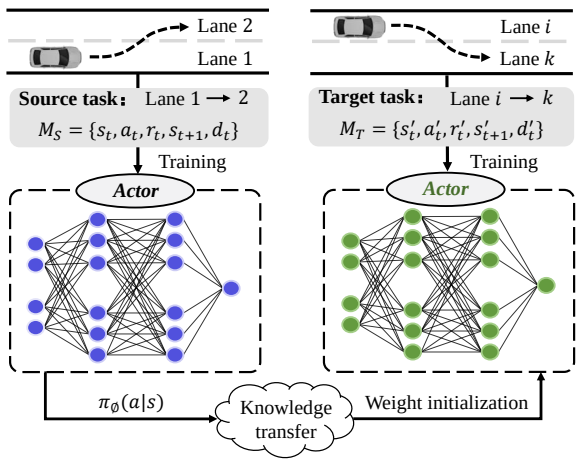

Fig. 3. Training architecture of the TTL-based trajectory planning method.

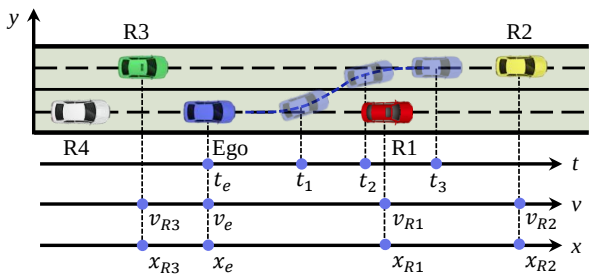

Fig. 4. Analysis of the vehicle motion process. The important parameters at key moments are marked in the figure.

The actor network is updated periodically through gradient descent, as shown below:

$$\nabla_\phi J(\phi) = N^{-1} \sum \nabla_a Q_{\theta_1}(s,a)|_{a=\pi_\phi(s)} \nabla_\phi \pi_\phi(s) \tag{10}$$

Finally, soft update techniques are used to synchronize the target network parameters with the main network.

Due to the similarity among lane change scenarios across different lanes, TL is employed to accelerate the convergence of models for planning different lane change trajectories. Fig. 3 illustrates the training architecture of the proposed TTL method, which intuitively reflects the knowledge transfer mechanism between the source task network (lane change from lane 1 to lane 2) and the target task network (lane change from lane i to lane k, where $i \neq k$). The parameters of the target task network are initialized using the weights of the source task network.

B. Safety assessment module

Unlike previous learning-based methods, this subsection introduces the VMSA-based safety assessment module to enhance safety interpretability.

1) Motion process analysis: As shown in Fig. 4, Ego's lane change is affected by the leading (R1) and following (R4) vehicles in the current lane, and the leading (R2) and following (R3) vehicles in the target lane. To analyze the safety constraints, the lange change process is divided by several key moments: t_e is when the decision is made, t_1 and t_2 mark the start and completion of entering the target lane, and t_3 is when the lane change is complete.

$$t_i = t_e + \Delta t_i, \quad i \in \{1,2,3\} \tag{11}$$

where Δt_i is the time interval, which is obtained from the lane change trajectory loaded by the motion planning module. The motion planning module generates a trajectory library based on different initial speeds, as shown in Fig. 1.

According to the motion process, the collision avoidance conditions with risky vehicles are analyzed:

- R1: From t_e to t_2, Ego remains in the original lane and should stay behind R1 for safety.
- R2: During lane change (t_1 to t_3), Ego enters the target lane and should avoid colliding with R2. After changing lanes, it should adjust speed to maintain a safe distance. Thus, risk periods include the lane change phase and post-lane change distance adjustment.
- R3: From t_1 to t_2, Ego is in the target lane and may collide with R3. After t_2, R3 reacts using the IDM model. Ego should stay ahead of R3 during $t_1 < t < t_2$, and ensure the gap at t_2 meets braking requirements.
- R4: It follows Ego using the IDM model. Its influence can be ignored.

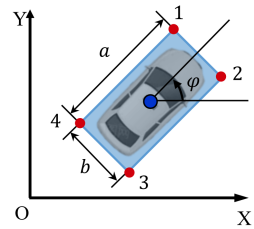

Fig. 5. The vehicle is modeled as a rectangle to determine the occupied space.

2) Safety constraints formulation: Fig. 5 establishes a vehicle rectangular model in the plane coordinate system. The longitudinal direction of the road is the x-axis, and the vertical direction is the y-axis. The coordinates of the vehicle's center are denoted as (x, y), and the coordinates of the four corner points can be calculated as follows:

$$\begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ x_4 & y_4 \end{bmatrix} = \begin{bmatrix} x & y \\ x & y \\ x & y \\ x & y \end{bmatrix} + \frac{1}{2} \begin{bmatrix} a\cos\varphi - b\sin\varphi & a\sin\varphi + b\cos\varphi \\ a\cos\varphi + b\sin\varphi & a\sin\varphi - b\cos\varphi \\ -a\cos\varphi + b\sin\varphi & -a\sin\varphi + b\cos\varphi \\ -a\cos\varphi - b\sin\varphi & -a\sin\varphi + b\cos\varphi \end{bmatrix} \tag{12}$$

where a and b are the length and width of the vehicle, respectively. φ is the heading angle.

The motion trajectories of the surrounding vehicles can be predicted using kinematics [37]. Based on the analysis of vehicle motion, the variations of the longitudinal positions of Ego, R1, R2, and R3 over time are illustrated in Fig. 6. The green area in the figure is the security zone, representing the safe relative positions where Ego and the risky vehicle will not collide. The time parameters $t_i, i \in \{1,2,3\}$ can all be obtained from t_e and the motion planning module, as shown in Eq. (11).

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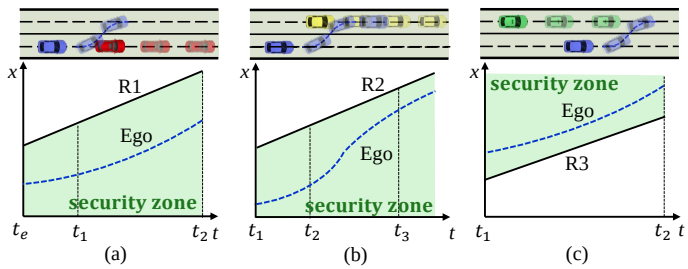

Fig. 6. Motion process diagram of Ego and the risky vehicle.

In the period $t_e < t < t_2$, there is a risk of collision between Ego and R1, as shown in Fig. 6(a). Under the guidance of the reward term r_{speed} in Eq. (6), the policy tends to select actions with higher speed, leading Ego to increase speed within safety constraints during the lane change process. The safety constraint for Ego with R1 only needs to be satisfied at time t_2, as shown below:

$$\begin{aligned} &f(x_e + \Delta x_{t_2},\ y_e + \Delta y_{t_2},\ \varphi_{t_2}) < \\ &\quad g(x_{R1} + v_{R1}\Delta t_2 - d_s,\ y_{R1},\ \varphi_{R1}) \end{aligned} \tag{13}$$

where $f(x, y, \varphi)$ and $g(x, y, \varphi)$ represent the maximum and minimum longitudinal coordinates of the vehicle's contour points x_1, x_2, x_3, x_4, as defined in Eq. (12). $\Delta x_{t_2}, \Delta y_{t_2}$ are the longitudinal and lateral distance traveled by Ego from time t_e to t_2, obtained from the trajectory library, as shown in Fig. 1. φ_{t_2} is the heading angle of Ego at time t_2. Let x_i, y_i ($i \in \{e, R1, R2, R3\}$) denote the positions of Ego and surrounding vehicles at t_e, and φ_{R_i} ($i \in \{1, 2, 3\}$) the constant heading of R_i under IDM. $d_s = a_{max} \cdot \max\{\Delta t_2\}$ is a safety margin for speed variation during prediction and is set to 2 m.

The collision avoidance between Ego and R2 depends on two factors: the lane change trajectory from t_1 to t_3 and the following distance required after the lane change. As shown in Fig. 6(b), the safety constraint is as follows:

$$\begin{aligned} &f(x_e + \Delta x_{t_2} + d_0,\ y_e + \Delta y_{t_2},\ \varphi_{t_2}) < \\ &\quad g(x_{R2} + v_{R2}\Delta t_2 - d_s,\ y_{R2},\ \varphi_{R2}) \\ &\text{s.t.} \quad d_0 = \begin{cases} d_{\text{IDM}}, & \text{if } v_{t_3} > v_{R2} \\ 0, & \text{else} \end{cases} \end{aligned} \tag{14}$$

where d_{IDM} is the required braking distance, and v_{t_3} is the speed of the Ego at time t_3.

Fig. 6(c) analyzes the position relationship between Ego and R3. At t_1, if Ego's speed and position are both greater, the distance will increase, meeting lane change requirements. Otherwise, a safety distance dsafe is needed. At t_2, a proper following distance d_0 should be maintained to avoid a rear-end collision. Therefore, the safety constraint with R3 is expressed as follows:

$$\begin{aligned} &\begin{cases} g(x_e + \Delta x_{t_1},\ y_e + \Delta y_{t_1},\ \varphi_{t_1}) > \\ \qquad f(x_{R3} + v_{R3}\Delta t_1 + d_s,\ y_{R3},\ \varphi_{R3}),\ \text{if } v_{t_1} > v_{R3} \\ g(x_e + \Delta x_{t_1},\ y_e + \Delta y_{t_1},\ \varphi_{t_1}) > \\ \qquad f(x_{R3} + v_{R3}\Delta t_1 + d_{safe} + d_s,\ y_{R3},\ \varphi_{R3}) \\ \quad \text{and} \quad g(x_e + \Delta x_{t_2},\ y_e + \Delta y_{t_2},\ \varphi_{t_2}) > \\ \qquad f(x_{R3} + v_{R3}\Delta t_2 + d_0 + d_s,\ y_{R3},\ \varphi_{R3}),\ \text{else} \end{cases} \\ &\text{s.t.} \quad d_{safe} = (v_{R3} - v_e)(t_2 - t_1) \\ &\qquad d_0 = \begin{cases} d_{IDM}, & \text{if } v_{R3} > v_{t_2} \\ 0, & \text{else} \end{cases} \end{aligned} \tag{15}$$

where v_{t_1} and v_{t_2} represent the speed of the Ego vehicle at time t_1 and t_2, respectively.

By solving the Eqs. (13)(14)(15), the required safety domain for a successful lane change can be determined. It is important that the parameters related to the Ego's lane change process can be obtained through the trajectory planning module. By pre-planning the trajectory, potential safety issues caused by unknown or uncertain parameters are avoided.

C. Lane selection module

Intelligent vehicles need to consider lane changes when encountering obstacles during driving. This study models the lane change decision problem as a MDP and addresses it using a value-distribution-based DRL approach.

1) Dueling Double DQN: DQN suffers from overestimation in Q-value prediction, which may lead to suboptimal action selection. To address this issue, Hasselt *et al.* [38] introduced the Double DQN algorithm. The improved target Q-value is as follows:

$$y^{DDQN} = r + \gamma Q(s', \arg\max_{a'} Q(s', a'; w); w') \tag{16}$$

where w and w' denote the parameters of the main network and the target network, respectively.

In certain cases, actions have less impact on Q-values than states. Dueling DQN separates state value and action advantage, then sums them to compute Q-values.

$$\begin{aligned} Q(s, a; \theta, \omega, \tau) =& V(\text{s};\theta, \tau) \\ &+ \left(A(s, a; \theta, \omega) - \frac{1}{|\text{A}|} \sum_{a'} A(s, a'; \theta, \omega) \right) \end{aligned} \tag{17}$$

where V and τ represent the state value and its network parameters, respectively, while A and w represent the advantage of the state-action pair and its network parameters. $|\text{A}|$ is the dimension of the action space.

Combining the Q-value calculation approach of Dueling DQN, the Q network in Double DQN is split into the sum of the state value network and the action advantage network, reducing overestimation and improving decision-making. This improved network is called Dueling Double DQN (D3QN). Due to the advantages of D3QN [39], it is used in the lane selection module for discrete action decision-making.

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Algorithm 1 Integrated Policy with VMSA

Initialize decision distance threshold d_m
for $t = 0$ **to** $MaxSteps$ **do**
  Observe environment information s_t, and the distance d_1 to the leading vehicle R1
  Set decision output $action = 0$, execute command $execution = 0$, waiting lane change $w_gap = False$
  if Ego vehicle goes straight and $d_1 < d_m$ and $w_gap = False$ **then**
    if obvious environmental changes **then**
      Select $action \sim$ Network(s_t) # Learning-based
      # Verifying if safety constraints are met according to Eqs. (13)(14)(15)
      Set $execution \sim$ VMSA$(action)$
      if safety constraints are not met **then**
        # Straight driving as an alternative trajectory
        Set $w_gap = True$
      end if
    end if
  else if $w_gap = True$ **then**
    Set $execution \sim$ VMSA$(action)$
    if safety constraints are met **then**
      # Lane change decision is allowed
      Set $w_gap = False$
    end if
  end if
end for
# Perform operation
if $execution \neq 0$ **then**
  Execute lane change based on motion planning module
else
  Execute following based on motion planning module
end if

2) RL element: For the state space, the observed traffic flow is divided into three parts. First, the state of ego vehicle is described using lane k, speed v, and acceleration a, represented as $s_{ego} = (k, v, a)$. Second, the state of risk vehicles, including R1, R2 and R3, is denoted as S_r. To intuitively represent the relative position of risk vehicle r to the ego vehicle, its state is described using the longitudinal distance Δx_r, lane k_r, speed v_r, and acceleration a_r, i.e., $(\Delta x_r, k_r, v_r, a_r)$. Finally, the state of vehicles within 150 m ahead is represented as S_f, enabling the ego vehicle to make lane selection decisions based on global traffic characteristics. Each vehicle's state is represented as $(\Delta x_f, k_f, v_f)$.

The state space S can be expressed as:

$$S = (s_{ego}, S_r, S_f) \tag{18}$$

The action space is set as follows:

$$A = (-1, 0, 1) \tag{19}$$

where 0 represents staying in the current lane, -1 and 1 indicate left and right lane changes, respectively.

TABLE I
SIMULATION PARAMETERS

	Parameter	Value
Simulation	Simulation step	0.1 s
	Acceleration range during car-following	-4~2 m/s^2
	Acceleration range during lane change	-2~2 m/s^2
	Speed range	7~18 m/s
	Decision distance threshold	25 m
	Vehicle width/length	1.8/3 m
	Lane width	3.2 m
Trajectory planning network	Total training epochs	3000
	Replay memory size	1e6
	Actor update delay	2
	Learning rate actor/critic	3e-3/3e-3
	Discount factor	0.9
	Batch size	128
	Optimizer	Adam
Lane selection network	Total training epochs	6500
	Replay memory size	1e3
	Learning rate	1e-3
	Discount factor	0.99
	Batch size	32
	Optimizer	Adam

The reward function is designed to guide the agent in learning effective policies to increase speed. The design principles are as follows:

- Stability: avoiding frequent lane changes.
- Efficiency: encouraging a high average speed to improve driving efficiency.

At each moment during the lane selection network training, the reward received by the controlled vehicle can be described as:

$$\begin{aligned} & r_l = r_{\text{speed}} + r_{\text{stability}} + r_{\text{final}} \\ & \text{s.t.} \\ & \quad r_{\text{speed}} = w_7 \frac{v - v_{min}}{v_{max} - v_{min}} \\ & \quad r_{\text{stability}} = \begin{cases} -w_8, & \text{if lane changes} \\ 0, & \text{else} \end{cases} \\ & \quad r_{\text{final}} = \begin{cases} w_9 \frac{v_{avg} - v_{min}}{v_{max} - v_{min}}, & \text{if done} \\ 0, & \text{else} \end{cases} \end{aligned} \tag{20}$$

where v_{avg} is the average speed. r_{speed} is a small immediate reward that encourages the vehicle to accelerate. r_{final} is the final reward used to evaluate the overall performance of the episode. The weights w_7 and w_8 are set to 0.5 and 1 respectively to balance the rewards, while w_9 is set to 100 to accelerate model convergence.

3) Integrated policy: Algorithm 1 shows the integrated decision-making and motion planning policy. The policy follows a clear priority hierarchy:

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TABLE II
TRAINING RESULTS OF DIFFERENT METHODS

Method	Average reward	Difference	Average maneuvering steps	Difference	Average lateral deviation (m)	Difference
TD3	81.3	—	27	—	1.5	—
TTL	**96.1**	**+18.2%**	**22**	**-18.5%**	**1.1**	**-26.7%**

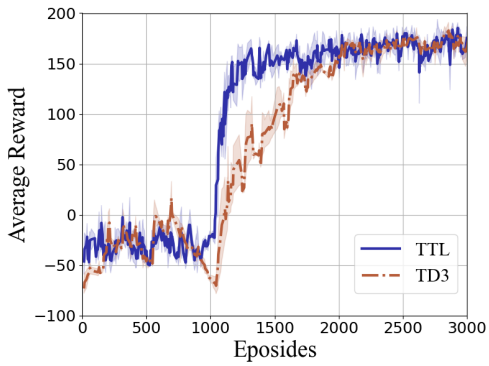

Fig. 7. Reward curve of the trajectory planning network. The shaded areas show the 90% confidence intervals across 3 seeds.

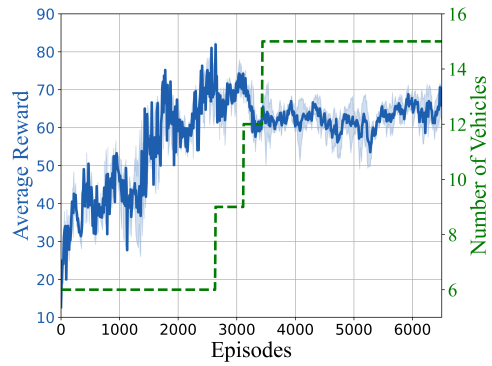

Fig. 8. Reward curve of the lane selection network. The shaded areas show the 90% confidence intervals across 3 seeds.

- First priority: the lane selection module, which determines whether to change lanes and selects the optimal target lane.
- Second priority: the safety assessment module, which verifies whether spacing constraints are satisfied.
- Third priority: the motion planning module, which executes car following or lane change maneuvers based on decision and safety assessment results.

When Ego first reaches the decision distance threshold, or when the leading vehicle in an adjacent lane changes, the event is regarded as an obvious change in the traffic environment. Under this framework, the lane selection module is activated to determine a target lane. The lane change command is then evaluated by the VMSA. Only when the safety constraints are satisfied does the motion planning module execute the lane change maneuver.

Specifically, the VMSA serves purely as a safety filter, blocking unsafe lane changes that violate safety distances requirements (considered planning failures). Therefore, it does not introduce any decision conflict or execution deadlock. When constraints aren't met (set $w_gap = True$), it executes a straight-driving trajectory as a safety alternative. Only when safety is ensured (set $w_gap = False$) does the vehicle switch to the optimal lane to enhance efficiency.

Training in difficult tasks is challenging and often unsuccessful. Curriculum learning is used to ensure stable training by gradually increasing task difficulty, allowing the agent to learn progressively and achieve better outcomes.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

To validate the performance of the proposed framework, this section presents the simulation setup and results analysis. Decision commands are issued whenever the controlled vehicle's distance to the leading vehicle is less than the decision distance threshold and traffic conditions change.

In training, the environment is reset and randomly initialized to enhance robustness. To enrich lane change behavior, The ego vehicle's speed ranges from [7-18] m/s, and surrounding vehicle's speed ranges from [7-13] m/s. When the preset conditions are met, the number of vehicles increases (+3 vehicles). The detailed parameters are shown in Table I.

In testing, three kinds of scenarios are used to evaluate the proposed framework. First, in a two-lane scenario (randomly spaced static obstacle and insufficient lane change gap), the focus is on verifying the collision avoidance performance without considering lane selection. Then, a three-lane dynamic traffic flow scenario is used to comprehensively validate both lane selection and collision avoidance performance.

A. Training results

For the trajectory planning network, the source task involves lane change from the right lane to the left, trained using TD3. While the target task (left-to-right) uses a combination of TD3 and TL. In Fig. 7, the source task reward curve converges after 2000 episodes, indicating effective policy learning. In contrast, the target task reward curve shows convergence at 1200 episodes, demonstrating that TL accelerates training. Table II shows the average training performance of the entire process. The TTL algorithm enhances both model generalization and real-time capability.

Fig. 8 presents the lane selection network's training results. In the early stages of training, the agent makes random lane change decisions. As the parameters are optimized, the agent learns the correct policy. The number of vehicles on the road

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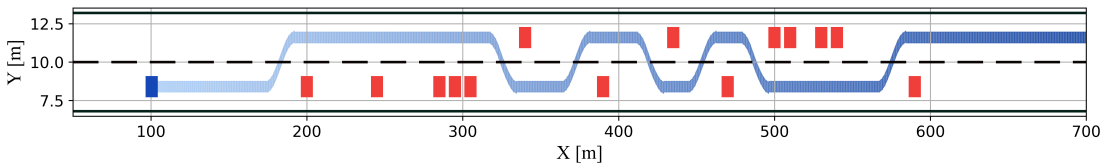

Fig. 9. The driving trajectory of agent in Scenario 1.

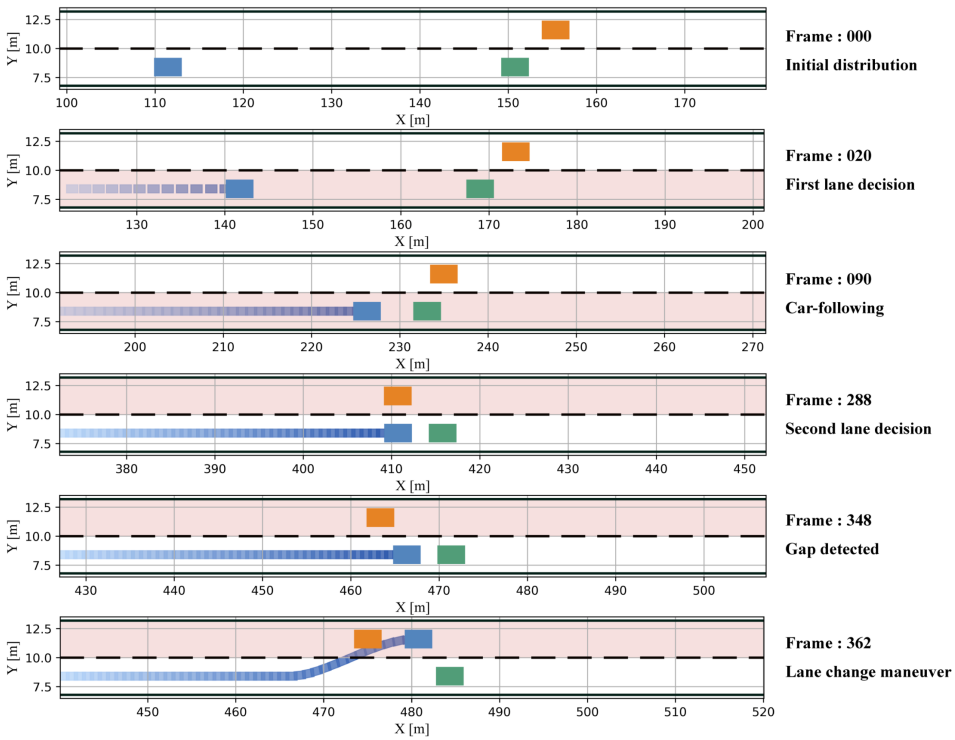

Fig. 10. Trajectory results of the agent in Scenario 2. The proposed framework effectively determines the appropriate lane, avoids frequent lane changes, and continuously monitors the gap for timely and safe lane changes.

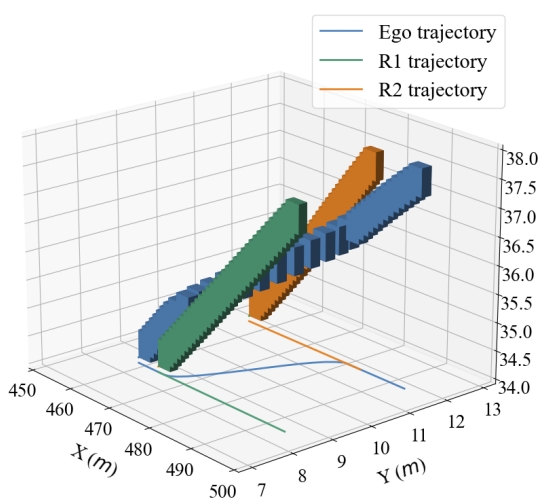

Fig. 11. Visualization of 3D trajectory.

increases from 6 to 15, and the reward tends to converge. The oscillation is caused by the randomness of the scenarios.

B. Scenario 1: randomly spaced static obstacle scenario

In this scenario, static vehicles are selected as obstacles to evaluate the proposed framework. This common simulation set-up uses static obstacles to generate a higher frequency of lane changes [33], effectively testing the performance of the lane change maneuver.

This scenario involves 15 vehicles randomly distributed along a 500-meter road. The controlled vehicle should identify the necessary lane change moments and pass the obstacle section with the fewest lane changes. The trajectory is shown in Fig. 9. The controlled vehicle is initialized at 100 m and detects an obstacle at 175 m. Based on the lane selection network, it chooses the adjacent lane and performs a lane change. Between 200-300 m, the controlled vehicle does not make unnecessary lane changes. The trajectory shows that the vehicle only changes lanes when necessary and can safely and smoothly pass the obstacle section.

C. Scenario 2: insufficient lane change gap scenario

The major limitation of traditional lane change methods is that the execution of the planning module can lead to failures and even collisions if wrong decisions are made. A scenario

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TABLE III
COMPARISON WITH BENCHMARK METHODS FOR 6 VEHICLES (AVERAGE OVER 10 RUNS)

Method				Average velocity (m/s)	Average acceleration (m/s^2)	Average lane changes	Average collisions	Average curvature ($10^{-3}/m$)
	Lane selection	Motion planning	Safety enhancement					
M1	*Keep*	—	—	10.48 ± 2.01	—	—	—	—
M2	*MOBIL*	*SUMO*	—	14.72 ± 2.44	0.15 ± 0.54	4.20 ± 1.30	*0*	6.17 ± 2.45
M3	*D3QN*	*TTL*	*DRL penalty*	16.72 ± 1.13	0.10 ± 0.47	2.50 ± 1.90	0.40 ± 0.70	1.58 ± 1.66
M4	*D3QN*	*SUMO*	*VMSA*	16.58 ± 1.27	0.16 ± 0.57	2.60 ± 1.07	0.20 ± 0.42	6.18 ± 3.18
Proposed	***D3QN***	***TTL***	***VMSA***	$\mathbf{16.70 \pm 1.13}$	$\mathbf{0.13 \pm 0.50}$	$\mathbf{1.90 \pm 0.57}$	***0***	$\mathbf{1.40 \pm 0.48}$

TABLE IV
COMPARISON WITH BENCHMARK METHODS FOR 15 VEHICLES (AVERAGE OVER 10 RUNS)

Method				Average velocity (m/s)	Average Acceleration (m/s^2)	Average lane changes	Average collisions	Average curvature ($10^{-3}/m$)
	Lane selection	Motion planning	Safety enhancement					
M1	*Keep*	—	—	11.13 ± 2.53	—	—	—	—
M2	*MOBIL*	*SUMO*	—	14.19 ± 3.19	0.07 ± 0.48	9.50 ± 4.48	*0*	7.53 ± 2.82
M3	*D3QN*	*TTL*	*DRL penalty*	14.79 ± 3.52	0.10 ± 0.50	3.50 ± 1.43	1.00 ± 1.41	1.24 ± 0.46
M4	*D3QN*	*SUMO*	*VMSA*	15.82 ± 2.55	0.14 ± 0.54	3.20 ± 1.48	0.50 ± 0.53	3.65 ± 1.82
Proposed	***D3QN***	***TTL***	***VMSA***	$\mathbf{16.10 \pm 2.34}$	$\mathbf{0.11 \pm 0.47}$	$\mathbf{2.90 \pm 1.45}$	***0***	$\mathbf{1.21 \pm 0.70}$

with two lanes and three vehicles is set up, as shown in Fig. 10. The blue vehicle is controlled using the proposed framework, while the other two vehicles use the IDM model to drive along the current road.

In the initial phase, the controlled vehicle detects that the distance is less than the decision threshold and accelerates at maximum rate.

At frame 20, the distance reaches the decision threshold. The decision network chooses to stay in the current lane, as the vehicle ahead is faster than the one in the adjacent lane. The controlled vehicle adjusts accordingly.

At frame 288, the ID of the vehicle ahead in the adjacent lane changes, indicating a change in traffic conditions. The decision network issues a command to switch lanes to overtake. Meanwhile, the VMSA checks the distance. Until frame 348, the driving behavior switches from following to lane change.

The 3D trajectory of the lane change process is shown in Fig. 11. The trajectory does not show any conflicts, indicating that the motion planning module makes reasonable decisions.

D. Scenario 3: comprehensive validation in three-Lane scenarios

Tables III and IV present the testing results of the proposed framework in three-lane dynamic traffic flow scenarios. The number of vehicles follows the curriculum learning setup, increasing sequentially from 6 to a maximum of 15. During testing, each vehicle's initial lane and maximum speed are random. Four benchmark methods (M1, M2, M3 and M4) are set for comparison. M1 means the Ego keeps going straight in the current lane without lane change. M2 adopts MOBIL and SUMO to simulate lane change behavior, representing a common autonomous driving control framework [25, 32]. To analyze the impact of VMSA, M3 applies DRL with a collision penalty (if collision: $r = -20$). M4 is used to evaluate the effectiveness of the TTL module compared to SUMO.

In lane selection, compared to M1's straight-driving strategy and M2's MOBIL method that only considers adjacent vehicles, the proposed DRL-based framework dynamically adjusts decisions based on traffic distribution, selecting the most efficient lane. This improves average speed while reducing lane changes.

In motion planning, the TTL algorithm generates multi-objective optimized lane change trajectories with lower average curvature and acceleration than SUMO's (M4) default trajectories.

In terms of safety, the DRL used in M3 cannot handle unseen scenarios during training, leading to collisions in stochastic environments. In contrast, the proposed VMSA method enforces safety constraints to ensure safe lane change gaps, providing stronger robustness. The required parameters of VMSA are provided by the trajectories planned by TTL, as shown in Eq. (11). As a result, collisions occur in M4 due to parameter mismatches between SUMO trajectories and VMSA.

Overall, the proposed framework effectively guides the controlled vehicle to suitable lanes in various complex scenarios, improving driving speed.

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TABLE V
PERFORMANCE COMPARISON OF LANE CHANGE METHODS

Method	Lane change	H.I. risk	Success rate
Li *et al.*, 2024 [40]	1	No	100%
Tang *et al.*, 2022 [32]	2	No	84–100%
Saxena *et al.*, 2020 [41]	> 1	Yes	90%
Li *et al.*, 2022 [33]	> 1	Yes	88–94%
Proposed	$\mathbf{> 2}$	**Yes**	**100%**

Table V aims to evaluate the safety advantages by comparing the results of the 15-vehicle experiment in Scenario 3 with previous studies. A High Interaction Risk (H.I. risk) scenario refers to a situation in which the ego vehicle is surrounded by both front and rear vehicles in both its current and target lanes, posing a potential risk of collision. The 100% collision-free performance is achieved by the VMSA module, which corrects unsafe decisions and ensures that lane change gaps meet safety requirements. Note that all results are obtained under the assumptions described in Section II.

E. Computation time

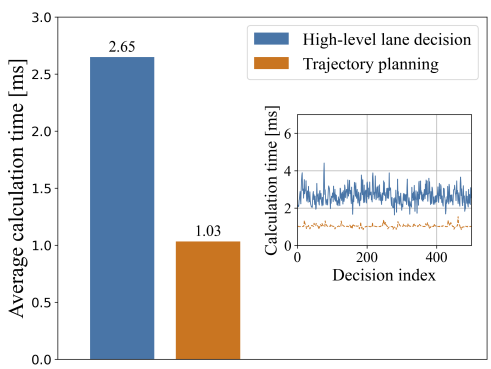

Fig. 12. The computation time of the modules involving action output in the proposed framework.

TABLE VI
COMPUTATION TIME COMPARISON OF DIFFERENT METHODS

	Decision-making		Motion planning	
	Method	Time (ms)	Method	Time (ms)
Liao *et al.*, 2024 [25]	DRL	0.42	MPC	24.24
Li *et al.*, 2024 [40]	POMDP	20.00	Convex optimization	15.00
Proposed	**DRL**	**2.65**	**DRL**	**1.03**

The proposed framework exhibits excellent real-time performance. In the dynamic traffic flow scenario, the computation times for each module are shown in Fig. 12. The lane selection network requires approximately 3 ms. The trajectory planning network takes about 1 ms due to its smaller observation space.

Table VI presents a comparison of computation times with related literature [25, 40]. Although there are differences in experimental setups, the results still demonstrate that the proposed method achieves millisecond-level computation, highlighting its significant advantage in real-time performance.

V. CONCLUSION

To address the challenge that learning-based and rule-based methods cannot simultaneously ensure both safety and real-time performance, an integrated autonomous driving decision-making and motion planning framework with collision avoidance and real-time capabilities is proposed. The integrated framework has the following advantages: i) a safety assessment module based on vehicle kinematics analysis provides interpretable safety guarantees. ii) DRL and TL approach enable the generation of multi-objective optimized trajectories while ensuring real-time performance. iii) a DRL policy based on global traffic features is used for lane selection, which improves driving speed. The hypothesis is supported by the following results: i) safety ensured: the proposed framework achieves zero collisions across all tested scenarios (see Tables III–V). ii) real-time enhancement: both decision-making and motion planning are executed with millisecond-level computation time (see Table VI), outperforming the MPC-based method. iii) driving efficiency improvement: the vehicle achieves higher average velocity and fewer lane changes, while maintaining lower average acceleration and curvature (see Tables III–IV).

The proposed framework is analyzed in mixed traffic environments with both human-driven and intelligent vehicles. However, some limitations remain in the current analysis. First, signal errors caused by noise are not considered. Second, to reduce the complexity of multi-vehicle interaction modeling, the proposed lane change policy is only applied when surrounding vehicles are assumed to move straight. Otherwise, the controlled vehicle follows the IDM to move straight and avoid uncertain collisions, as stated in the assumptions in Section II-A. Future work will focus on improving the framework. Potential enhancements include modeling active lane change interactions between vehicles and incorporating dynamic noise to improve real-world performance.

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